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ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B. Tech (Aerospace Eng.)

Semester: III

Name of the Paper: Aero Thermodynamics

Paper Code: **TAS 303**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Describe the following terms: system, surrounding, boundary, and universe.	CO1
(b)	State the first law of thermodynamics. Also, give the expressions of energy balance for the following non-flow processes: i) cyclic process and ii) process undergoing a change of state.	
(c)	Describe the concept of continuum.	
Q2	(20 marks)	
(a)	Describe the steady flow energy equation, and apply it to a compressor.	CO2 & CO3
(b)	A stationary mass of gas is compressed without friction from an initial state of 0.5 m ³ and 0.12 MPa to a final state of 0.25 m ³ and 0.12 MPa. If there is a transfer of 40 kJ of heat from the gas during the process, evaluate the change in internal energy of the gas.	
(c)	A machine operating as a heat pump extracts heat from the surrounding atmosphere, is driven by a 6 kW motor and supplies 3000 kJ/min heat to a room for its heating. Find COP of this heat pump. How this COP will be affected if the same machine is used to cool the room requiring 3000 kJ/min of heat rejection?	
Q3	(20 marks)	
(a)	What is the entropy of a system? Derive the relation for entropy change in an irreversible process.	CO4
(b)	State the second and third laws of thermodynamics. What is a PMM2?	
(c)	Describe the availability and the irreversibility.	
Q4	(20 marks)	
(a)	Explain the working of an IC engine with the help of corresponding Diesel cycle on p-V and T-s diagrams.	CO5
(b)	Describe the Brayton cycle with its efficiency.	
(c)	Draw the p-v and T-s diagrams for the Atkinson and Stirling cycles. Also write the expressions for their efficiencies.	
Q5	(20 marks)	
(a)	Explain the working of a refrigeration cycle with its different components.	CO6
(b)	What are different components of a steam power plant, explain with the governing thermodynamic cycle.	
(c)	Describe the saturated and superheated steam and write the expression for the dryness fraction.	

